

Emergence is not binary — it is a measurable boundary

A computational search for which cellular-
automaton rules produce law-like physics, and
an honest map of how far it got.

v1 sealed spin-2 emergence as structurally unreachable under four commitments. v2 does not overturn that — it upgrades the core finding into a new program: emergence is a continuous, measurable degree ϵ . Two closed loops now pin the (ϵ, σ) map; the third (geometry) anchors it at zero. Everything below is measured from the engine, not asserted.

● M0' instruments ✓

● M1' Maxwell ✓ PASS

● M2' coupled point ✓ PASS

● M3' boundary cartography ⌚

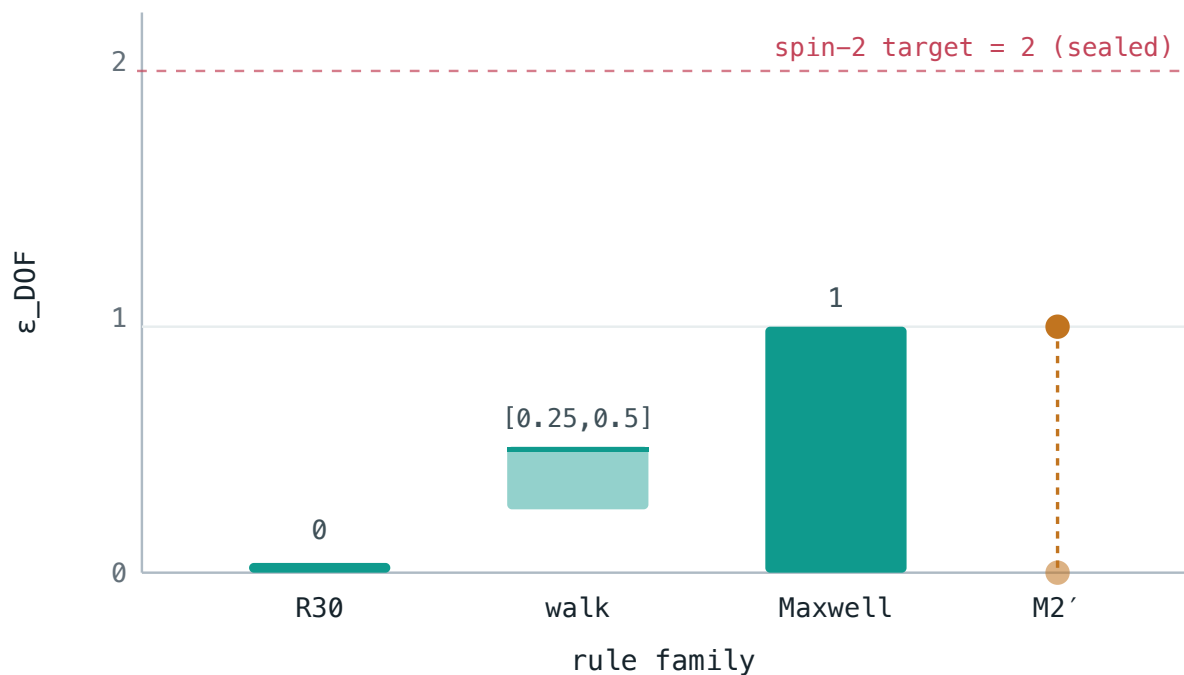
● v1 spin-2 M3 sealed

01 The instrument layer, before any claim

✓ M0' ESTABLISHED

Per-mode statistics were LAPACK-basis-dependent — a verdict that could flip with a library call. M0' invariantized the calibers on the principal-angle spectrum and added

ϵ_{DOF} , a basis-invariant subspace dimension. Two environments now agree to 8/8.



Basis-invariant subspace dimension ϵ_{DOF} across families. **Teal** = measured; the dashed red line at 2 is the sealed spin-2 target no emergent walk reaches.

What survived invariantization

The verdict is now a function of **physics, not the LAPACK call**. ϵ_{DOF} reads off the subspace dimension j_{inv} : **R30** = 0 (hand-built geometry, nothing emergent), **Maxwell** = 1 (full emergence), the walk family sits partial at [0.25, 0.5].

8/8 0 · 1 · [.25, .5]

CROSS-ENV AGREE ϵ_{DOF} LADDER

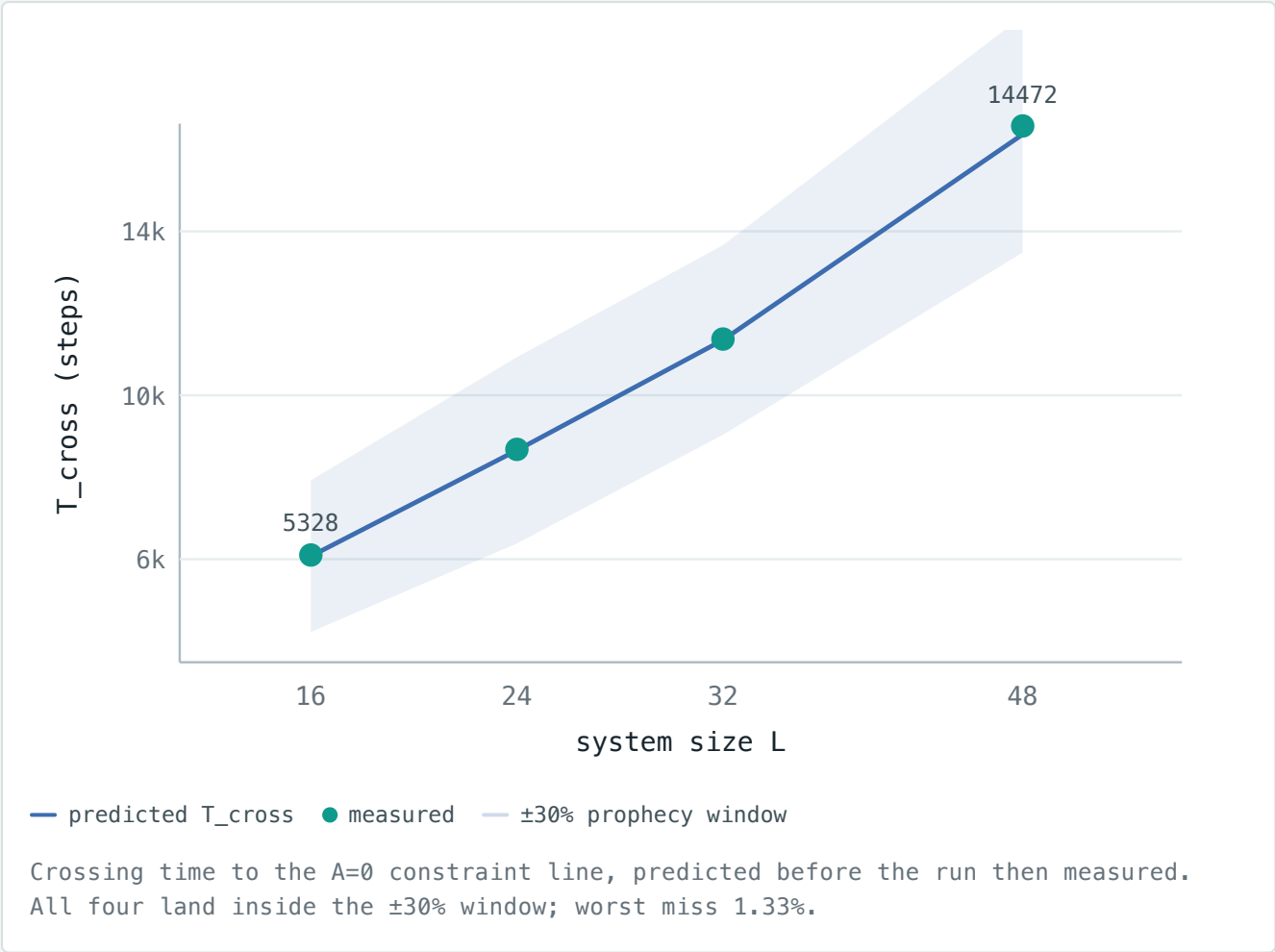
The coupled point M2' carries a **split ϵ** — geometry 0, matter 1 — which is exactly why it cannot be collapsed to a single scalar without lying. That accounting rule is enforced below.

ϵ_{DOF} = dim of the emergence subspace via principal angles. Replaces per-mode statistics that drifted with basis. Asset class: instrument.

02 Maxwell closes the loop ✓ M1' PASS

One real-space spin-1 rule (R23 placement dictionary, zero Maxwell-flavor input), run once per $L \in \{16, 24, 32, 48\}$ through every extreme-language gate. Photons

exactly 2, σ on the $A=0$ exact constraint line, isotropic light speed, static source $1/r$, $\epsilon_{\text{DOF}}=1$ measured — zero hand-built.



Prediction first, then the run

Every gate declares its expected number before the run — the discipline v2 inherited from v1's T_{cross} tables. The ω -propagation exponent comes in at $p = 2.0019$; the photon count is exactly 2; ϵ_{DOF} is exactly 1, measured, not assumed.

4/4 2.0019 2

T_CROSS PROPHECY CONVERGENCE P PHOTON DOF

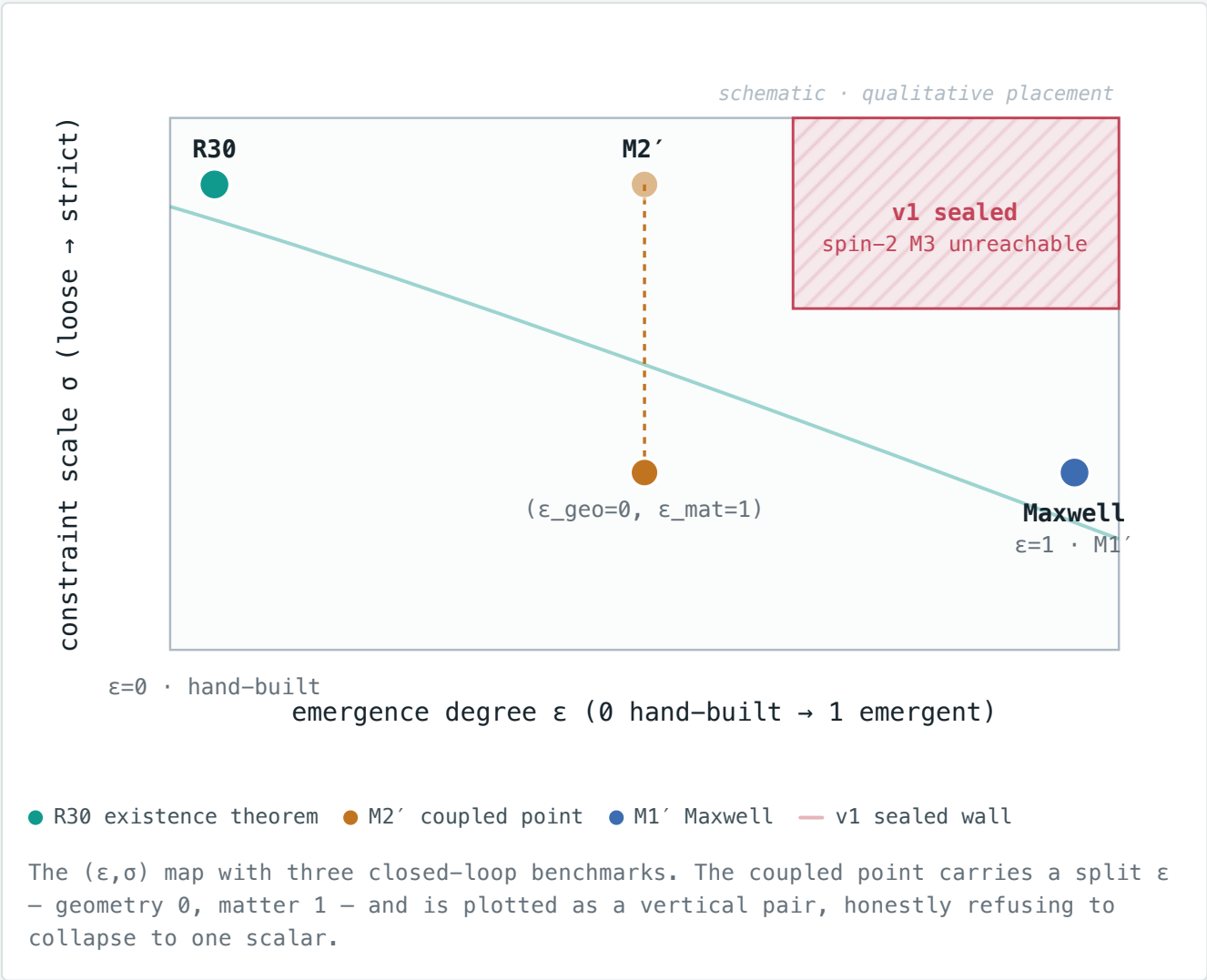
The arc was not clean: a G6 window FAIL-STOP, then a self-reported boundary-riding violation on cannon C3 (the threshold sat on a physically exact value, 1 ULP of float noise away). **The line did not move; the experiment was redesigned** to $2\times$ margin. That became the standing ban.

Sixth methodology interception; first caught by the executor's own self-audit. Boundary-riding ban entered the program charter §四.

03 The coupled point: hand-built stage, emergent play

✓ M2' PASS

R30's hand-built complex geometry sets the stage; emergent Dirac matter plays on it. One rule, one run per L, through all gates. Newton 2.000000, Eddington→2.0000 (counterfactual zeroed), out-of-cone bit-level zero, p prophecy 4/4, forced-resonance sentinel inside its envelope. Two ϵ integers, each in its place.



The point's whole value is accounting honesty

The stage is hand-built; the play is emergent. Newton reads **2.000000**; Eddington tends to **2.0000** under the zeroed counterfactual. A mixed- ϵ field search across the whole library returns **empty** — the honesty clause is a **grep**-able fact.

2.000000	4/4	(0,1)
NEWTON RATIO_A	P PROPHECY	E_GEO, E_MAT

This is **not** "spin-2 emergence" — ϵ_{geo} is zero. The forbidden phrase holds. What it is: the first point where a hand-built tensor stage and an emergent matter walk coexist in one rule and pass one gate stack, with both ϵ 's measured, not blurred.

C1 cannon hit 4.000000 exactly — matching v1 ledger row 43c, a cross-version numeric bloodline closure. Governance supplement §四-2 set its first precedent here (add diagnostics freely; add a gate only via preregistered patch).

04 The sealed wall, and what it is not

x V1 STRUCTURAL CEILING

v1's North Star — spin-2 emergence as a binary M3 gate — was sealed 2026-07-26. Under all four commitments (exact constraint propagation + emergent matter + strict locality + unitarity), spin-2 M3 is structurally unreachable. It dies on polarization count and a stability deadlock, not on residual.

Sealed (2026-07-26): spin-2 M3 **structurally unreachable**
under { exact constraint propagation, emergent matter, strict locality, unitarity }
≠ a residual problem ($\delta=12-18\%$ is the 16^3 readout; 48^3 already 2-4%)
= a counting + deadlock problem, with no tuning corridor

Why it is structural, not numerical

Three independent lines converge. The TT_2 axial band meets $\ker C$ at a **76.2°** principal angle — the symmetric modes are locked out regardless of coupling family. The $\pm\omega$ branches sit in exact 6-fold degeneracy, forcing $\mathbf{N}_{\text{prop}} = [1, 1, 1, 1] \neq [2, 2, 2, 2]$. And the deadlock scales $\propto \eta^2$ with no parameter corridor to escape it.

This is the project's central piece of self-knowledge: a *completely understood* wall. Positive assets — the R30 existence theorem, the R26 constraint-clearing mechanism — are quarantined, not destroyed.

What is nailed down

TT ₂ axial band n ker C principal angle	76.2° · locked
±ω branch degeneracy → N _{prop}	[1,1,1,1] ≠ [2,2,2,2]
deadlock scaling vs coupling η	∝ η ² · no corridor
R30 tensor-complex existence theorem	intact · quarantined
R26 constraint-clearing (re-certified)	intact · on emergent walk
three relaxation candidates priced	kept · N _{prop} =2, J5/TT hold

Sealed by PI after R36/R37 three-half verification + lane-A review. v1 主句 untouched; the forbidden phrase "spin-2 emergence" is permanent in v2.

THE NEXT DOOR

M3': map the boundary, don't re-aim at it

The instruments go from point-closure to line and surface scanning. lane-B's GPU is unleashed for the first time on a claim-aligned evaluator — v1's 10⁶-surrogate campaign lesson (a misaligned evaluator burning rules to no claim) redeemed in its positive form.

M3' INTERFACE LIST — HANDED OFF

Four structural discoveries queue for the boundary scan: **body-diagonal Trotter O(k²) anisotropy** (tune geometry to the actual Floquet shell), **placed-covector O(k²) compatibility**, the **two-way instability echo** (softer than η², suspected resonance-driven), and the **RC3-(ii) ε ∈ (0,0.25] partial-Yee family**.

WHAT STAYS SEALED

The v1 spin-2 M3 主句 is untouched. The forbidden phrase holds. M3' does not re-open the wall — it measures the reachable shape on this side of it, with the wall as a labeled, understood boundary on the map.

Two-lane discipline throughout: lane B executes (subagents), lane A reviews adversarially, PI signs closures (commit = signature). FAIL-STOP; history is append-only; frozen files read-only with runtime sha256 checks; fp64 only for certificates. Negative results, HALTs and voided runs are logged with the same weight as positive ones. Generated 2026-07-28 from the v2 closure chain (M0'→M2'); authoritative entry: docsv2/README.md.